

Caleb Kyei

Control & Power Systems Engineer · Triple-Master's Student (DREAM)
Barcelona · LinkedIn

SUMMARY

Power electronics and control engineer with **4.5 years at Signify R&D** (2 as an engineer), including a digital PID design tool for high-power PFC that cut design time from **a week to minutes**. Now in the Erasmus Mundus DREAM master's (Nantes, Berlin, Barcelona), focused on converter and grid control: weak-grid converter control at CITCEA-UPC, d-q current control of a 10 kVA grid-following inverter, MPC for a 1.55 MW EV powertrain, and day-ahead solar forecasting for 50Hertz (**2nd of 16** on RMSE). Seeking a 6-month Master's thesis, 1 February – 31 July 2027.

EXPERIENCE

Signify (formerly Philips Lighting) **Eindhoven, NL**

Electronics Development Engineer Aug 2023 – Aug 2025

- Developed switch-mode LED drivers for luminaire builders, including soft-switching power stages and isolated flyback auxiliary supplies; horticulture lighting among the applications.
- Worked across embedded hardware/software and driver communication interfaces (e.g. Modbus).

Graduation Intern (Digital Control) Feb 2023 – July 2023

- Built a MATLAB/Simulink, Python and Excel design tool for a digital PID controller in high-power PFC, automating parameter generation, optimisation and verification; design time cut from a week to minutes. **(Grade: 9)**

Power Electronics Intern Feb 2022 – Jan 2023

- Investigated SiC MOSFET power stages for outdoor LED drivers and embedded buck-stage switching control. **(Grade: 8)**

Test Automation Intern Feb 2021 – Jan 2022

- Automated embedded testing of emergency LED drivers in R&D.

Robotics Team Pi (FRC) **Eindhoven, NL**

Software Designer, then Mentor Sept 2020 – Aug 2022

- Wrote C++ autonomous aim-assist and swerve/differential drive control; later led the team's software mentoring.

RESEARCH INTERESTS

- Converter & inverter control
- MMC & grid-forming control
- Model predictive & optimal control
- EV powertrains & FOC drives
- Power system stability
- ML forecasting for energy systems

SKILLS

- MATLAB, Simulink, Stateflow
- LTspice
- Python (PyTorch, scikit-learn, pvlib)
- Reinforcement Learning
- Power Electronics & Inverters
- Model-based Design (MBD)
- Docker, GitLab CI/CD, Git
- C, C++

LANGUAGES

- English (Native/Bilingual)
- Dutch (Intermediate/B1)
- French (Elementary/A2+)
- German (Beginner/A1.2)

CERTIFICATIONS

- Agile with Atlassian Jira
- Performance Coach
- Advanced Modeling & Simulation of Power Electronic Systems using PLECS (Plexim, 2023)

HONORS & AWARDS

- Top Talent Scholarship, Fontys (4×), for academic excellence
- ECHO mentorship programme (Google mentor)

EDUCATION

Master's Degrees (Triple-Degree Program) **2025 – 2027**

École Centrale de Nantes Sep 2025 – Feb 2026
Master's degree, Control and Robotics *(Semester 1)*
Focus: Adv. Control Architectures & Power System Dynamics
Top grade: Dynamic Components of Power Systems (A+)

- **Robust & Optimal Control:** Designed non-linear control strategies (SMC, LQR); implemented Kalman Filters and System ID for state estimation.
- **Optimization & Grid Dynamics:** Applied MILP/KKT scheduling optimization and modeled power system stability/inertia via DAE.
- **Project – 2MW DFIG Wind Turbine:** Designed a supervisory FSM to orchestrate MPPT vs. rated-power modes with anti-chattering hysteresis, coordinating RSC torque with pitch actuators; validated stability and C_p optimization in MATLAB/Simulink under turbulent wind.

HTW Berlin Mar 2026 – Aug 2026
Master's degree, Electrical Engineering *(Semester 2)*
Focus: Electric Vehicle Drives, Grid Control, Power Electronics, Wind & Photovoltaic Engineering
Top grades: Power Electronics (1.0), Electric Vehicle Motors (1.3); German scale, 1.0 best

- **EV Powertrain, MPC & ACC:** Modelled a 1.55 MW quad-motor EV drive in Simulink with PMSM FOC and a state-space MPC speed controller, plus adaptive cruise control (ISO 15622 constant time gap, closing-rate damping); collision-free in all six aggressive-lead cases, speed overshoot below 0.05 km/h.
- **Probabilistic Grid Forecasting (50Hertz):** Built a Python day-ahead solar forecast for the 50Hertz area (ENTSO-E): physics-based PV ensemble with spread-adaptive quantiles. RMSE under 6% of the 11.6 GW mean daily peak; 2nd of 16, behind only the TSO baseline, from public data only.
- **Power Electronics & Inverters:** Modelled a 10 kVA grid-following VSI in Simulink (team project): SRF-PLL, decoupled d-q PI current control with feedforward (IMC-tuned, 1 kHz bandwidth), SPWM and analytical LCL filter design; sized a 3 MW PV boost stage from ripple specifications.
- **Renewables Analytics:** Resolved PV inverter identification via correlation fingerprinting (SonnJA! facility) and modeled wind turbine aerodynamics (Betz limit, torque/pitch control).

Universitat Politècnica de Catalunya Sep 2026 – Aug 2027
Master's degree, Energy Engineering *(Semesters 3 & 4)*
Focus: Grid-Forming Control, Renewable Grid Integration, Power Quality, AI & Data Science

- **Project – Grid-Connected Converter Control (in progress, CITCEA-UPC):** Built an averaged grid-following VSC with SRF-PLL and decoupled d-q current control; swept short-circuit ratio and X/R to assess stability versus PLL bandwidth; implemented DC-link voltage control, current limiting and anti-windup.
  - Grid-forming control of a modular multilevel converter (MMC) is a planned extension, to be completed before the thesis.
- **Coursework (Semester 3, in progress):** Renewable Grid Integration (grid-forming, grid codes, HVDC, microgrids); Power Quality; AI in Energy Systems (MLP, LSTM, CNN); Data Science for Energy Systems.
- **Semester 4:** Master's Thesis, 6 months, 1 February – 31 July 2027.

Bachelor's Degree

Fontys University of Applied Sciences, Eindhoven, NL 2019 – 2023
BSc Electrical and Electronics Engineering, *Cum Laude*

- Thesis: 'Digital Control in Power Systems' (9.0). Fontys PROUD Honours Programme (Chair). **Minor:** Performance Coaching and Training.